

Overview of materials for PVC, Molded

Categories: [Polymer](#); [Thermoplastic](#); [Vinyl \(PVC\)](#); [PVC, Molded](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "PVC, Molded". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	0.480 - 2.50 g/cc	0.0173 - 0.0903 lb/in ³	Average value: 1.29 g/cc Grade Count:629
Volatile Organic Compounds (VOC) Content	0.300 g/l	0.300 g/l	Average value: 0.300 g/l Grade Count:9
Water Absorption	0.0200 - 0.400 %	0.0200 - 0.400 %	Average value: 0.278 % Grade Count:12
Particle Size	63.0 - 250 µm	63.0 - 250 µm	Average value: 157 µm Grade Count:12
Viscosity	0.680 - 2.27 cP	0.680 - 2.27 cP	Average value: 1.48 cP Grade Count:11
Viscosity Test	101 - 145 cm ³ /g	1.01 - 1.45 dl/g	Average value: 119 cm ³ /g Grade Count:10
Viscosity Number	79.0 - 180 cm ³ /g	0.790 - 1.80 dl/g	Average value: 121 cm ³ /g Grade Count:14
Thickness	1000 - 3000 microns	39.4 - 118 mil	Average value: 1100 microns Grade Count:26
Linear Mold Shrinkage	0.00200 - 0.0500 cm/cm	0.00200 - 0.0500 in/in	Average value: 0.00579 cm/cm Grade Count:29
Melt Flow	2.00 - 25.0 g/10 min	2.00 - 25.0 g/10 min	Average value: 12.5 g/10 min Grade Count:7
Spiral Flow	63.5 - 94.0 cm	25.0 - 37.0 in	Average value: 83.0 cm Grade Count:6
ASTM Color	-32.4 - 97.2	-32.4 - 97.2	Average value: 18.7 Grade Count:56
Mechanical Properties	Metric	English	Comments
Hardness, Shore A	28.0 - 99.0	28.0 - 99.0	Average value: 71.0 Grade Count:506
Hardness, Shore D	34.0 - 87.0	34.0 - 87.0	Average value: 68.0 Grade Count:71
Hardness, Shore C	75.0 - 95.0	75.0 - 95.0	Average value: 85.0 Grade Count:5
Tensile Strength, Ultimate	1.47 - 59.0 MPa	213 - 8560 psi	Average value: 16.6 MPa Grade Count:594
Tensile Strength, Yield	29.6 - 52.0 MPa	4290 - 7540 psi	Average value: 43.2 MPa Grade Count:23
Elongation at Break	3.00 - 600 %	3.00 - 600 %	Average value: 312 % Grade Count:566

Modulus of Elasticity	0.00163 - 3.24 GPa	0.237 - 470 ksi	Average value: 2.14 GPa Grade Count:50
Flexural Yield Strength	2.83 - 103 MPa	410 - 15000 psi	Average value: 74.8 MPa Grade Count:42
Flexural Modulus	1.65 - 3.31 GPa	239 - 480 ksi	Average value: 2.67 GPa Grade Count:39
Izod Impact, Notched	0.300 - 28.4 J/cm	0.562 - 53.2 ft-lb/in	Average value: 6.36 J/cm Grade Count:42
	0.534 - 7.47 J/cm @Temperature -40.0 - 0.000 °C	1.00 - 14.0 ft-lb/in @Temperature -40.0 - 32.0 °F	Average value: 3.11 J/cm Grade Count:4
	0.534 - 7.47 J/cm @Thickness 3.18 - 3.18 mm	1.00 - 14.0 ft-lb/in @Thickness 0.125 - 0.125 in	Average value: 3.11 J/cm Grade Count:4
Izod Impact, Unnotched	0.400 - 11.3 J/cm	0.749 - 21.3 ft-lb/in	Average value: 4.42 J/cm Grade Count:16
Tensile Impact Strength	63.0 - 168 kJ/m²	30.0 - 79.9 ft-lb/in²	Average value: 139 kJ/m² Grade Count:7
Tear Strength Test	145 - 530	145 - 530	Average value: 308 Grade Count:64
Tear Strength	15.0 - 75.0 kN/m	85.6 - 428 pli	Average value: 33.6 kN/m Grade Count:23
Dart Drop Total Energy	534 - 2140 J/cm	1.00 - 4.00 ft-lb/mil	Average value: 1660 J/cm Grade Count:7

Electrical Properties	Metric	English	Comments
Electrical Resistivity	2.00e+6 - 1.00e+15 ohm-cm	2.00e+6 - 1.00e+15 ohm-cm	Average value: 1.04e+14 ohm-cm Grade Count:19
	>= 2.00e+10 ohm-cm @Temperature 70.0 - 70.0 °C	>= 2.00e+10 ohm-cm @Temperature 158 - 158 °F	Average value: 1.05e+12 ohm-cm Grade Count:5
Comparative Tracking Index	400 - 600 V	400 - 600 V	Average value: 575 V Grade Count:4
Hot Wire Ignition, HWI	60.0 - 120 sec	60.0 - 120 sec	Average value: 93.3 sec Grade Count:8

Thermal Properties	Metric	English	Comments
Maximum Service Temperature, Air	40.0 - 100 °C	104 - 212 °F	Average value: 52.2 °C Grade Count:29
Deflection Temperature at 0.46 MPa (66 psi)	66.0 - 90.6 °C	151 - 195 °F	Average value: 77.3 °C Grade Count:18
Deflection Temperature at 1.8 MPa (264 psi)	62.0 - 87.2 °C	144 - 189 °F	Average value: 71.2 °C Grade Count:48
Vicat Softening Point	68.0 - 91.0 °C	154 - 196 °F	Average value: 77.3 °C Grade Count:27
Softening Point	160 - 190 °C	320 - 374 °F	Average value: 174 °C Grade Count:7
Minimum Service Temperature, Air	-57.5 - -10.0 °C	-71.5 - 14.0 °F	Average value: -34.0 °C Grade Count:19
Brittleness Temperature	-72.0 - -3.00 °C	-97.6 - 26.6 °F	Average value: -36.4 °C Grade Count:187

Decomposition Temperature	185 - 200 °C	365 - 392 °F	Average value: 188 °C Grade Count:5
UL RTI, Electrical	50.0 - 80.0 °C	122 - 176 °F	Average value: 56.0 °C Grade Count:5
UL RTI, Mechanical with Impact	50.0 - 70.0 °C	122 - 158 °F	Average value: 54.0 °C Grade Count:5
UL RTI, Mechanical without Impact	50.0 - 80.0 °C	122 - 176 °F	Average value: 56.0 °C Grade Count:5
Flammability, UL94	HB - 5VA	HB - 5VA	Grade Count:57
Oxygen Index	25.0 - 34.0 %	25.0 - 34.0 %	Average value: 28.4 % Grade Count:27

Optical Properties	Metric	English	Comments
Haze	5.00 - 9.00 %	5.00 - 9.00 %	Average value: 6.86 % Grade Count:7
Transmission, Visible	66.0 - 90.0 %	66.0 - 90.0 %	Average value: 86.1 % Grade Count:56

Processing Properties	Metric	English	Comments
Melt Temperature	154 - 213 °C	310 - 415 °F	Average value: 179 °C Grade Count:119

Component Elements Properties	Metric	English	Comments
Bromine, Br	0.0100 %	0.0100 %	Average value: 0.0100 % Grade Count:9
Cadmium, Cd	0.00500 %	0.00500 %	Average value: 0.00500 % Grade Count:9
Chromium, Cr	0.0100 %	0.0100 %	Average value: 0.0100 % Grade Count:9
Lead, Pb	0.0100 %	0.0100 %	Average value: 0.0100 % Grade Count:9
Mercury, Hg	0.0100 %	0.0100 %	Average value: 0.0100 % Grade Count:9

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.